Kaihua Qin

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CURRENT POSITION

Assistant Professor September 2025 – present
Department of Computer Science
University of Warwick

EDUCATION

PhD in Computer Science 2019 – 2024
Imperial College London London, United Kingdom
Advisor: Prof. Arthur Gervais
Thesis: Blockchain and Decentralized Finance Security: Incentives, Attacks, and Mitigations

MSc in Communications and Signal Processing 2014 – 2015
Imperial College London London, United Kingdom

BE in Information Engineering 2010 – 2014
Southeast University Nanjing, China

RESEARCH

My research focuses on understanding and tackling security challenges in emerging computational systems. I study how new system architectures create novel security, trust, and incentive problems, and develop principled methods to analyze and address them. Across these domains, I am interested in both uncovering new vulnerabilities and designing practical defenses, combining rigorous systems and security research with AI-based techniques for analyzing, protecting, and optimizing complex systems.

A major part of my work investigates decentralized systems such as blockchains and DeFi, drawing on distributed systems, applied cryptography, confidential computing, game theory, and mechanism design to understand their security and incentive structures. I also increasingly study AI-driven and agentic systems, including the security risks that arise when autonomous agents interact with software, users, markets, and one another. Ultimately, my goal is to understand the security foundations of emerging technologies and translate that understanding into practical tools and mechanisms that make future digital infrastructure more secure, trustworthy, and robust.

GRANTS AND AWARDS

- Warwick IAS Interdisciplinary Research Development Award (5K GBP) 2026
  Integrated Risk Modelling for Decentralized Finance
- Grants Innosuisse Start-Up Innovation Project (703K CHF) 2024 - 2026
  AI-Enhanced Smart Contract Auditing
- Ethereum Foundation Academic Grant (55K USD) 2024
  Training an EVM-native Generative Pre-trained Transformer
- The Latest in DeFi Research (30K USD) 2024
  Fellowship on stablecoin research

- Ethereum Foundation Academic Grant (50K USD) 2023
  Dynamic Analysis Framework for EVM
- Ethereum Foundation Academic Grant (72K USD) 2023
  Denial-of-Service Implications of Blockchain Censorship
- Ethereum Foundation Academic Grant (25K USD) 2023
  Blockchain Censorship — Quantitative Analysis of Censorship on Public Blockchains
- Cybersecurity Best Practical Paper Award 2023
- Best DeFi Papers Award 2023

PUBLICATIONS AND PREPRINTS

Bibliometrics from Google Scholar: citations = 3,877, h-index = 23, i10-index = 28 (as of October 2026).

32. **Qin, Kaihua**, Dawn Song, and Arthur Gervais. "SCDBench: A Benchmark for LLM-Based Smart Contract Decompilers". In *Advances in Neural Information Processing Systems*, vol. 39. 2026.

31. Zhou, Yifan, Jiang Xiao, and **Kaihua Qin**. "Security Limits of Mining Before Validation in Nakamoto Consensus". *arXiv preprint arXiv:2609.29222* (2026).

30. Yang, Sen, **Kaihua Qin**, Aviv Yaish, and Fan Zhang. "Insecurity Through Obscurity: Veiled Vulnerabilities in Closed-Source Contracts". In *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security*. ACM, 2026.

29. Wang, Ziyue, Cheuk Wang Maurice Ng, Chenchen Yu, Strick Sheng, **Kaihua Qin**, and Liyi Zhou. "Transferable Self-Evolving Playbooks for Agentic Security Auditing". *arXiv preprint arXiv:2606.16420* (2026).

28. Wang, Ziyue, Jiangshan Yu, **Kaihua Qin**, Dawn Song, Arthur Gervais, and Liyi Zhou. "TxRay: Agentic Postmortem of Live Blockchain Attacks". *arXiv preprint arXiv:2602.01317* (2026).

27. **Qin, Kaihua**, Zhe Ye, Zhun Wang, Weilin Li, Liyi Zhou, Chao Zhang, Dawn Song, and Arthur Gervais. "Enhancing Smart Contract Security Analysis with Execution Property Graphs". *Proceedings of the ACM on Software Engineering* 2, no. ISSTA (2025): 1101–1122.

26. Guan, Maggie Yongqi, Yaman Yu, Tanusree Sharma, Molly Zhuangtong Huang, **Kaihua Qin**, Yang Wang, and Kanye Ye Wang. "Security Perceptions of Users in Stablecoins: Advantages and Risks within the Cryptocurrency Ecosystem". In *2025 IEEE Symposium on Security and Privacy (SP)*, 2753–2771. IEEE, 2025.

25. Tsuchiya, Taro, Liyi Zhou, **Kaihua Qin**, Arthur Gervais, and Nicolas Christin. "Blockchain Amplification Attack". *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 9, no. 1 (2025): 1–30.

24. Potter, Yujin, Kornrapat Pongmala, **Kaihua Qin**, Ariah Klages-Mundt, Philipp Jovanovic, Christine Parlour, Arthur Gervais, and Dawn Song. "What Drives the (In) stability of a Stablecoin?" In *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 316–324. IEEE, 2024.

23. Wahrstätter, Anton, Jens Ernstberger, Aviv Yaish, Liyi Zhou, **Kaihua Qin**, Taro Tsuchiya, Sebastian Steinhorst, Davor Svetinovic, Nicolas Christin, and Mikolaj Barczentewicz. "Blockchain censorship". In *Proceedings of the ACM Web Conference 2024*, 1632–1643. 2024.

22. Gan, Rundong, Liyi Zhou, Le Wang, **Kaihua Qin**, and Xiaodong Lin. "Defialigner: Leveraging symbolic analysis and large language models for inconsistency detection in decentralized finance". In *6th Conference on Advances in Financial Technologies (AFT 2024)*, 7–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2024.

21. Yaish, Aviv, **Kaihua Qin**, Liyi Zhou, Aviv Zohar, and Arthur Gervais. "Speculative {Denial-of-Service} Attacks In Ethereum". In *33rd USENIX security symposium (USENIX Security 24)*, 3531–3548. 2024.

20. David, Isaac, Liyi Zhou, **Kaihua Qin**, Dawn Song, Lorenzo Cavallaro, and Arthur Gervais. "Do you still need a manual smart contract audit?" *arXiv preprint arXiv:2306.12338* (2023).

19. Wahrstätter, Anton, Liyi Zhou, **Kaihua Qin**, Davor Svetinovic, and Arthur Gervais. "Time to bribe: Measuring block construction market". *arXiv preprint arXiv:2305.16468* (2023).

18. Zhou, Liyi, Xihan Xiong, Jens Ernstberger, Stefanos Chaliasos, Zhipeng Wang, Ye Wang, **Kaihua Qin**, Roger Wattenhofer, Dawn Song, and Arthur Gervais. "Sok: Decentralized finance (defi) attacks". In *2023 IEEE Symposium on Security and Privacy (SP)*, 2444–2461. IEEE, 2023.

17. Feng, Ding, Rupert Hitsch, **Kaihua Qin**, Arthur Gervais, Roger Wattenhofer, Yaxing Yao, and Ye Wang. "Defi auditing: Mechanisms, effectiveness, and user perceptions". In *International Conference on Financial Cryptography and Data Security*, 320–336. Springer, 2023.

16. Wang, Zhipeng, Stefanos Chaliasos, **Kaihua Qin**, Liyi Zhou, Lifeng Gao, Pascal Berrang, Benjamin Livshits, and Arthur Gervais. "On how zero-knowledge proof blockchain mixers improve, and worsen user privacy". In *Proceedings of the ACM Web Conference 2023*, 2022–2032. 2023.

15. Gai, Yu, Liyi Zhou, **Kaihua Qin**, Dawn Song, and Arthur Gervais. "Blockchain large language models". *arXiv preprint arXiv:2304.12749* (2023).

14. Raun, Christoffer, Benjamin Estermann, Liyi Zhou, **Kaihua Qin**, Roger Wattenhofer, Arthur Gervais, and Ye Wang. "Leveraging machine learning for bidding strategies in miner extractable value (mev) auctions". *Cryptology ePrint Archive* (2023).

13. Yaish, Aviv, Maya Dotan, **Kaihua Qin**, Aviv Zohar, and Arthur Gervais. "Suboptimality in defi". *Cryptology ePrint Archive* (2023).

12. **Qin, Kaihua**, Stefanos Chaliasos, Liyi Zhou, Benjamin Livshits, Dawn Song, and Arthur Gervais. "The blockchain imitation game". In *32nd USENIX Security Symposium (USENIX Security 23)*, 3961–3978. 2023.

11. **Qin, Kaihua**, Jens Ernstberger, Liyi Zhou, Philipp Jovanovic, and Arthur Gervais. "Mitigating decentralized finance liquidations with reversible call options". In *International Conference on Financial Cryptography and Data Security*, 344–362. Springer, 2023.

10. Wang, Zhipeng, **Kaihua Qin**, Duc Vu Minh, and Arthur Gervais. "Speculative multipliers on defi: Quantifying on-chain leverage risks". In *International Conference on Financial Cryptography and Data Security*, 38–56. Springer, 2022.

9. **Qin, Kaihua**, Liyi Zhou, and Arthur Gervais. "Quantifying blockchain extractable value: How dark is the forest?" In *2022 IEEE Symposium on Security and Privacy (SP)*, 198–214. IEEE, 2022.

8. **Qin, Kaihua**, Liyi Zhou, Pablo Gamito, Philipp Jovanovic, and Arthur Gervais. "An empirical study of defi liquidations: Incentives, risks, and instabilities". In *Proceedings of the 21st ACM internet measurement conference*, 336–350. 2021.

7. Zhou, Liyi, **Kaihua Qin**, Antoine Cully, Benjamin Livshits, and Arthur Gervais. "On the just-in-time discovery of profit-generating transactions in defi protocols". In *2021 IEEE Symposium on Security and Privacy (SP)*, 919–936. IEEE, 2021.

6. Zhou, Liyi, **Kaihua Qin**, Christof Ferreira Torres, Duc V Le, and Arthur Gervais. "High-frequency trading on decentralized on-chain exchanges". In *2021 IEEE Symposium on Security and Privacy (SP)*, 428–445. IEEE, 2021.

5. **Qin, Kaihua**, Liyi Zhou, Yaroslav Afonin, Ludovico Lazzaretti, and Arthur Gervais. "CeFi vs. DeFi–Comparing Centralized to Decentralized Finance". *arXiv preprint arXiv:2106.08157* (2021).

4. Zhou, Liyi, **Kaihua Qin**, and Arthur Gervais. "A2mm: Mitigating frontrunning, transaction reordering and consensus instability in decentralized exchanges". *arXiv preprint arXiv:2106.07371* (2021).

3. **Qin, Kaihua**, Liyi Zhou, Benjamin Livshits, and Arthur Gervais. "Attacking the defi ecosystem with flash loans for fun and profit". In *International conference on financial cryptography and data security*, 3–32. Springer, 2021.

2. Janin, Simon, **Kaihua Qin**, Akaki Mamageishvili, and Arthur Gervais. "Filebounty: Fair data exchange". In *2020 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 357–366. IEEE, 2020.

1. **Qin, Kaihua**, Henryk Hadass, Arthur Gervais, and Joel Reardon. "Applying private information retrieval to lightweight bitcoin clients". In *2019 Crypto Valley Conference on Blockchain Technology (CVCBT)*, 60–72. IEEE, 2019.

PROFESSIONAL SERVICES

Program Committee Chairs

- ACM CCS Workshop on Decentralized Finance and Security 2024 (co-chair with Liyi Zhou)
- ACM CCS Workshop on Decentralized Finance and Security 2023 (co-chair with Fan Zhang)

Program Committee Members

- ACM Conference on Computer and Communications Security (CCS '24, '25, '26)
- IEEE Symposium on Security and Privacy (S&P '25, '27)
- USENIX Security Symposium (Security '25, '27)
- The Network and Distributed System Security Symposium (NDSS '25, '26, '27)
- IEEE European Symposium on Security and Privacy (EuroS&P '26)
- International Conference on Financial Cryptography and Data Security (FC '24, '25, '26, '27)
- ACM ASIA Conference on Computer and Communications Security (ASIACCS '27)
- The international conference on Advances in Financial Technologies (AFT '24, '25, '26)
- Nanyang Blockchain Conference (NBC '26)
- IEEE International Conference on Blockchain and Cryptocurrency (ICBC '25)
- ACM Workshop on Decentralized Finance and Security (DeFi '21, '22, '25, '26)
- International Workshop on Formal Methods for Blockchains (FMBC '26)
- Designing DeFi Conference (D^2 '26)

- Web3 Scholars Conference (WSC '26)
- Crypto Economics Security Conference (CESC '22)

Reviewer

- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- ACM Transactions on Privacy and Security (TOPS)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- ACM Transactions on the Web (TWEB)
- Distributed Ledger Technologies: Research and Practice (DLT)
- Blockchain: Research and Applications (BCRA)
- IEEE Access
- Future Generation Computer Systems
- Financial Innovation
- Electronic Commerce Research
- ACM CHI conference on Human Factors in Computing Systems (CHI '24, '26)
- IEEE International Conference on Distributed Computing Systems (ICDCS '23)
- The Conference on Web and Internet Economics (WINE '22)
- International Conference on Information Systems (ICIS '22)
- The Web Conference (WWW '21)

EXPERIENCE

Co-founder December 2022 – Oct 2026
Decentralized Intelligence AG

Affiliated Researcher February 2024 – August 2025
Yale University
Host: Prof. Fan Zhang and Prof. Zhong Shao

Visiting Researcher May 2022 – August 2022
University of California, Berkeley
Host: Prof. Dawn Song

Research Intern February 2022 – April 2022
Chainlink Labs

Software Engineer March 2016 – August 2018
Cisco

Updated October 2026